VisionScript: Image Captioning System

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***Abstract*—This paper presents VisionScript-ImageCaptioning - BLIP, a four-milestone vision-language pipeline for automatic image captioning and visual question answering (VQA). The project evolved from a prompt-only GPT-2 baseline on Flickr30k (Milestone 1), through CLIP-conditioned prefix projection using a trainable 3.9M-parameter PrefixProjection layer (Milestone 2), a rigorous four-way fine-tuning comparison evaluated with BLEU, METEOR, CIDEr, and ROUGE-L metrics on 200 samples (Milestone 3), and a live Gradio web application with multi-mode BLIP captioning and VQA deployed on NVIDIA A100 (Milestone 4). The central quantitative finding is that LoRA (r=8, approximately 300K trainable parameters) achieves the best BLEU (0.0537), METEOR (0.2064), and ROUGE-L (0.3028) among all fine-tuning methods, while BLIP leads on CIDEr (0.2030). The final system was deployed live on an NVIDIA A100-SXM4-40GB GPU.**

***Index Terms*—image captioning, BLIP, CLIP, GPT-2, LoRA, prefix tuning, visual question answering, Flickr30k, fine-tuning comparison**

I. Introduction

Automatic image captioning — generating natural language descriptions of visual content — sits at the intersection of computer vision and natural language processing, with applications spanning accessibility tools for visually impaired users, content indexing, social media, and human-robot interaction. Recent advances in vision-language pretrained models have substantially improved caption fluency and semantic accuracy.

VisionScript-ImageCaptioning - BLIP began with a research question stated in the project proposal: Can a CLIP + GPT-2 pipeline generate stylistically distinct captions for the same image using style prefix conditioning — and how does caption quality vary across prompt-only, embedding injection, and fine-tuning approaches? The project was later extended in Milestone 4 to add multi-mode paragraph captioning and interactive VQA, deployed as a live Gradio application.

The project was developed by Group 5 for the IE 7615 Neural Networks and Deep Learning course at Northeastern University, with equal 25% contributions. Aditya Kumar led training and evaluation including metric setup for BLEU, METEOR, CIDEr, and ROUGE-L. Meghana Rao led demo and report writing. Simran Abhay Sinha led data and preprocessing including the Flickr30k pipeline, caption filtering, embedding extraction, and Drive caching. Sushmitha Sudharsan led model architecture including CLIP and GPT-2 integration, style prompts, and the `generate_caption()` helper.

All experiments from Milestone 2 onward ran on Google Colab with an NVIDIA A100-SXM4-40GB GPU, with models and datasets cached to Google Drive for reproducibility.

II. Related Work

Image captioning evolved from template-based methods and CNN-RNN encoder-decoder architectures [4] to attention-augmented models. Xu et al. [5] showed that soft attention over spatial image features substantially improves alignment between visual content and generated words.

CLIP [1] introduced contrastive pretraining on 400 million image-text pairs. BLIP [2] advanced this through bootstrapped pretraining on filtered web data, achieving state-of-the-art on captioning and VQA benchmarks. GPT-2 [3] provides a strong autoregressive language prior for open-ended generation.

Prefix Tuning [8] freezes the base model and learns continuous soft prompt tokens prepended to the input. LoRA [9] injects low-rank decomposition matrices into transformer attention layers, reducing trainable parameters by orders of magnitude. VisionScript-ImageCaptioning - BLIP empirically compares both approaches against full fine-tuning and the BLIP baseline.

III. System Architecture and Methodology

A. Milestone 1 — Automatic Image Captioning Baseline

The baseline pipeline used the Flickr30k dataset [7] from HuggingFace (`clip-benchmark/wds_flickr30k`, `split=test`). A subset of 1,000 image-caption pairs was curated after filtering captions with fewer than 5 words. Images were converted to RGB and cached as `flickr30k_1k.pkl` on Google Drive.

Models used: CLIP ViT-B/32 for 512-dim L2-normalized embedding extraction cached as `embeddings.pt`; BLIP (`Salesforce/blip-image-captioning-base`) for unconditional caption generation; and the GPT-2 tokenizer for caption length analysis only.

BLIP generation parameters: `num_beams=4`, `max_new_tokens=50`, `early_stopping=True`. Style prompt prefixes defined in this milestone — *Art Critic*: "An art critic observes this painting:"; *Casual Visitor*: "A museum visitor says about this painting:"; *Children's Guide*: "A tour guide explains this painting to children:". In Milestone 1 these prompts were defined but CLIP embeddings were not yet injected into GPT-2.

Actual BLIP captions on 5 Flickr30k test images are shown in Table I.

Key observation: BLIP generates image-grounded captions out of the box. However, captions were occasionally imprecise — the taekwondo scene was described as a rug and the rooftop as sky, motivating embedding injection in Milestone 2.

Fig. 1. Milestone 1 — BLIP baseline captions on 5 Flickr30k images.

TABLE I
MILESTONE 1 — BLIP BASELINE CAPTION RESULTS

#	Image Content	BLIP Caption
1	Man with orange hat	"a man wearing a hat..."
2	Dog running by fence	"a white picket fence..."
3	Taekwondo on mat	"a red, blue and white rug..."
4	Group in snow	"a group of people standing in the snow..."
5	Workers on rooftop	"the sky is blue..."

B. Milestone 2 — CLIP Embedding Injection into GPT-2

The `PrefixProjection` module maps CLIP's 512-dim embedding to GPT-2's 768-dim space via `Linear(512→7680)+Tanh`, reshaped to `[B, 10, 768]`, producing 10 visual prefix tokens. Trainable parameters: 3,939,840. GPT-2 was fully frozen; only the projection layer was trained.

Environment: NVIDIA A100-SXM4-40GB GPU. Dataset: 1,000 image-caption pairs, 16 batches per epoch. Training configuration — Optimizer: AdamW (`lr=2e-4`, `weight_decay=1e-2`); Scheduler: CosineAnnealingLR; Batch size: 64; Max caption length: 32; Epochs: 20. Training loss: epoch 1 $\approx$ 6.8 $\rightarrow$ epoch 2 $\approx$ 5.1 $\rightarrow$ epoch 3 $\approx$ 3.3 $\rightarrow$ epoch 4 $\approx$ 2.5, converging to $\approx$ 1.9 by epoch 20.

Generation parameters: prompt prefix "a photo of"; `num_beams=5`; `max_new_tokens=60`; `min_new_tokens=15`; `no_repeat_ngram_size=3`; `length_penalty=1.5`. Three decoding strategies compared: Greedy (fast but repetitive), Beam Search (structured), and Nucleus Sampling `top_p=0.9` (most natural).

Actual caption samples are shown in Table II.

Additional demo outputs: Test 1 (girl in flower field) — CLIP+GPT-2 hallucinated "I love you" text on shirt; BLIP: "a little girl in a field of flowers." Test 2 (orange cat on car) — CLIP+GPT-2 produced generic dog collar output despite image showing a cat; BLIP: "an orange cat laying on the hood of a car.

The prefix projection layer learned a usable mapping. However, outputs showed hallucination, object repetition, and scene misidentification. BLIP outperformed CLIP+GPT-2 on most qualitative samples at this stage.

TABLE II
MILESTONE 2 — CLIP+GPT-2 VS. BLIP CAPTION COMPARISON (5 SAMPLES)

#	Original	CLIP+GPT-2	BLIP
1	Man with orange hat	"a man wearing a shirt with the words 'I'm a man'..."	"a man wearing a hat and glasses with a cell in his hand"
2	Dog running in garden	"a dog with a dog collar on its back..."	"a dog running in a grassy field with a white picket fence..."
3	Female karate kick	"a man with a helmet on his head..."	"a red, blue, and white rug."
4	Five snowmobile riders	"a man and a woman wearing a ski mask and a backpack..."	"a group of people standing in the snow."
5	Two men on roof	"a man and a woman sitting on a bench in front of a fire hydrant."	"a blue sky with a few white clouds."

C. Milestone 3 — Fine-Tuning Methods Comparison

Milestone 3 compared four strategies on 200 Flickr30k samples using BLEU, METEOR, CIDEr, and ROUGE-L. The four methods and their configurations are shown in Table III.

TABLE III
METHODS COMPARED IN MILESTONE 3

Method	Description	Params	Epochs
Prefix Tuning	PrefixProjection only; GPT-2 frozen	3.9M	20
LoRA	r=8, alpha=32, target c_attn, dropout=0.1	~300K	15
Full Fine-Tune	Entire GPT-2 end-to-end	124M	15
BLIP Baseline	blip-image-captioning-base; no training	247M	N/A

LoRA architecture: `PeftModelForCausalLM` with `c_attn` replaced by `lora.Linear` (base `Conv1D`

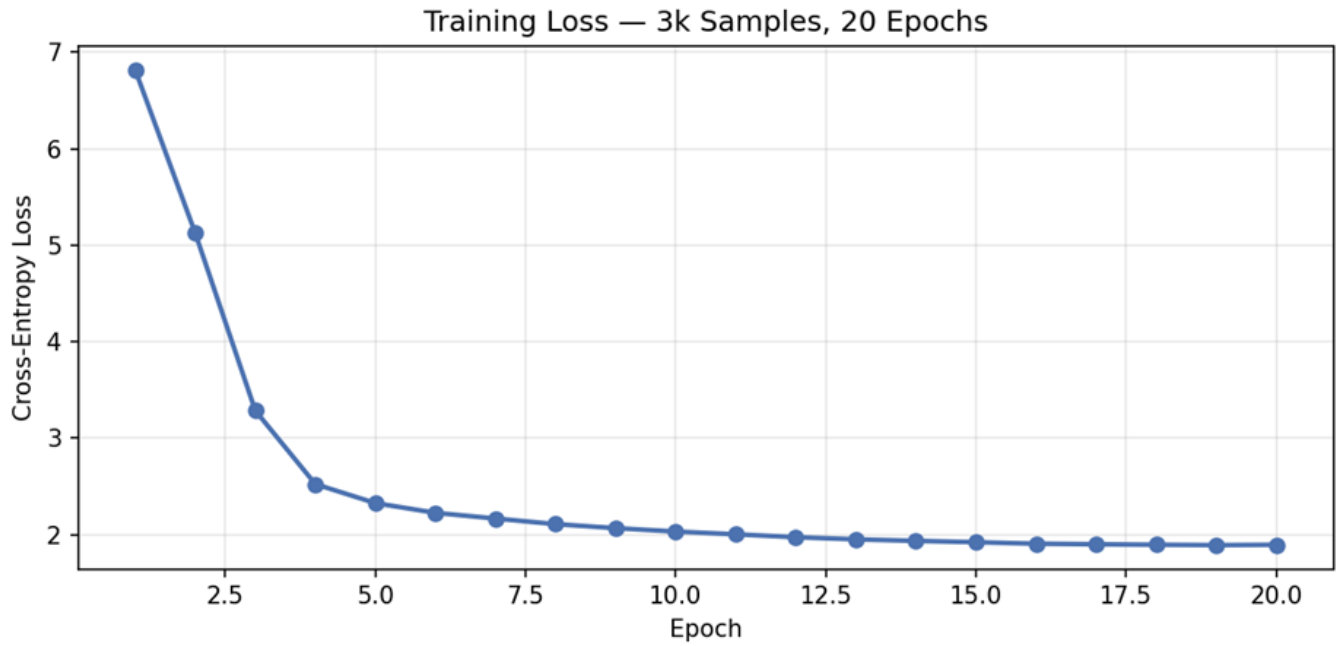

Fig. 2. Training loss — PrefixProjection, 20 epochs. Loss: 6.8 → 1.9.

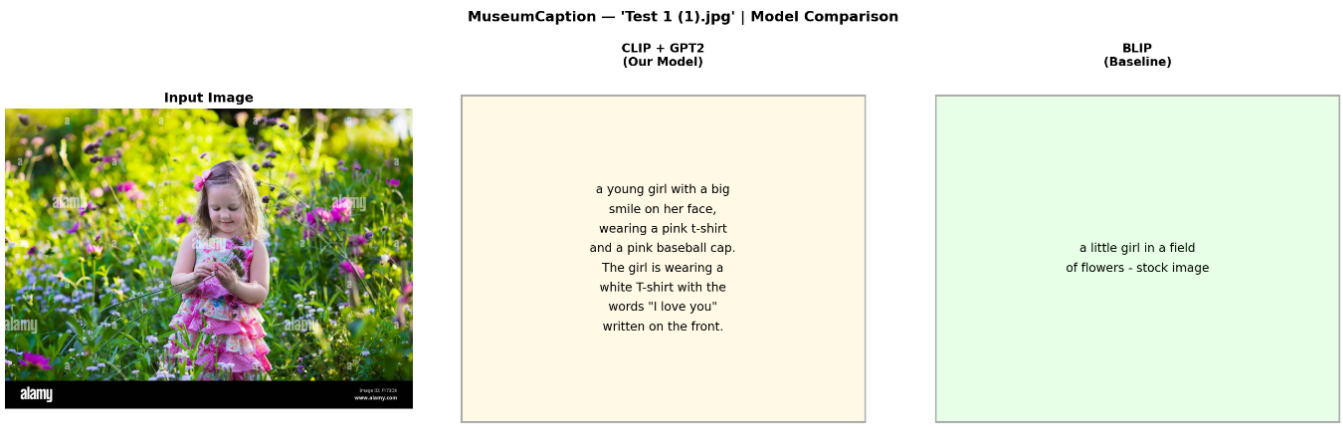

Fig. 3. M2 Test 1 — Girl in flower field. CLIP+GPT-2 hallucinated text; BLIP correct.

`nf=2304`, `nx=768`; `lora_A Linear(768→8)`; `lora_B Linear(8→2304)`; `dropout=0.1`).

Evaluation used `corpus_bleu` with `SmoothingFunction.method1` for BLEU; NLTK WordNet for METEOR; TF-IDF cosine similarity for CIDEr; `rouge_score` with `use_stemmer=True` for ROUGE-L. Exact quantitative results are shown in Table IV.

Beam width sensitivity on reference image (man with orange hat): bw=1: "a man with a beard and a beard"; bw=3: "a man with a beard and a mustache"; bw=5: "a man wearing a shirt with the words 'I'm a man' on it"; bw=7: "a man wearing a baseball cap with a baseball bat on his head"; bw=10: "a man wearing a baseball cap with a baseball bat on his head and a baseball glove on his arm."

Temperature sensitivity (nucleus sampling, same reference): temp=0.5: "a man with a beard"; temp=0.7: "a man with a mane of white hair"; temp=0.85: "a man wearing a baseball cap, a baseball bat and a baseball glove"; temp=1.0: "a man with his hands over his head, holding a bottle"; temp=1.2: "a man leaning on the back of a dog."

Decoding strategy comparison: Greedy/Prefix: "a man with a beard"; Greedy/LoRA: "a man in a hat and sunglasses with a red hat and glasses"; Greedy/Full FT: "a man in a red shirt and jeans sitting on a bench with a guitar"; Beam/LoRA: "a man wearing a hat and sunglasses with a smile on his face"; Beam/Full FT: "a man in a blue shirt and a woman in a red shirt standing in front of a building"; Nucleus/Prefix: "a man wearing a shirt with a cap on his head."

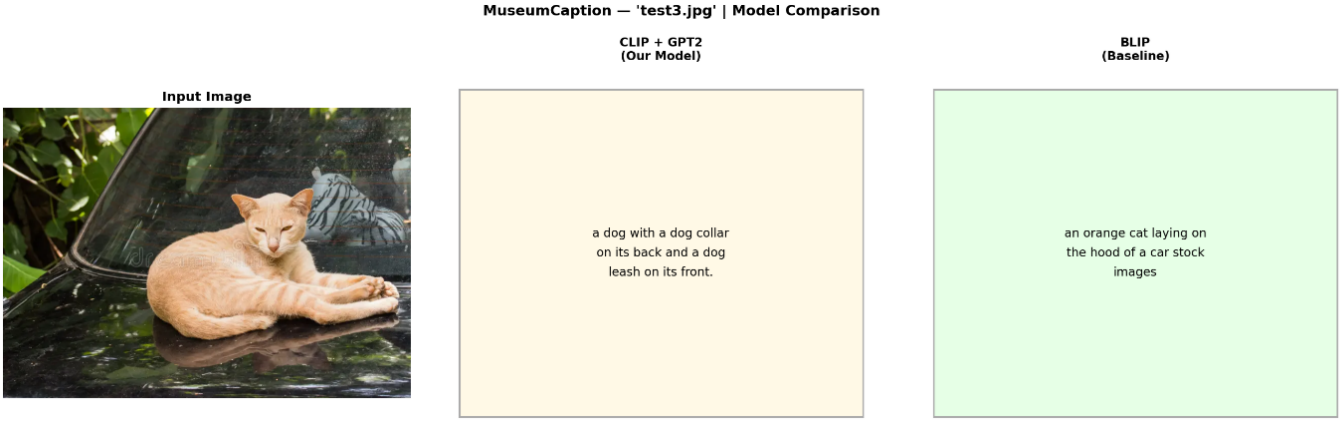

Fig. 4. M2 Test 2 — Orange cat on car. CLIP+GPT-2 misclassified; BLIP correct.

TABLE IV
MILESTONE 3 — QUANTITATIVE RESULTS (200-SAMPLE EVAL, FLICKR30K)

Model	BLEU	METEOR	CIDEr	ROUGE-L	Params
Prefix Tuning	0.0450	0.1768	0.1203	0.2601	3.9M
LoRA ⋆	**0.0537**	**0.2064**	0.1725	**0.3028**	∼300K
Full Fine-Tune	0.0474	0.1672	0.1002	0.2598	124M
BLIP ⋆	0.0417	0.1937	**0.2030**	0.3026	247M

⋆ = best score per metric.

Qualitative test (couple on bench in woods) — Prefix: long hallucinated output about T-shirt photo. LoRA: "a young woman sitting on a rock in the middle of the woods." Full FT: "a man in a blue shirt and a woman in a red shirt standing in front of a building." BLIP: "a man and a woman sitting on a bench in the woods." BLIP was most accurate. All GPT-2 variants failed on the lion test image, generating unrelated captions.

D. Milestone 4 — Multi-Mode Captioning and VQA

Two notebooks were produced. `Final_Prj2.ipynb` is a simple BLIP demo supporting image upload and URL input. Actual output on `IMG_9729.jpeg`: "a bouquet of red roses and white flowers on a table." `Final_Prj2NNDL.ipynb` is the full Gradio application deployed live at https://f5acdfd336db95b20c.gradio.live on NVIDIA A100-SXM4-40GB.

The pipeline is: Image → BLIP Encoder-Decoder → Caption / Answer Generation. Three modes: Single Caption (concise one-sentence); Paragraph Caption (detailed multi-sentence); VQA (question-conditioned answer). Generation parameters are shown in Table V.

Actual system outputs — Single (Lion): "a lion laying in the grass with a blue sky in the background." Paragraph (Lion): "a lion laying on the ground with its head turned to the side and it's looking at the camera, with a blue sky behind it in the background of a field of grass and a few trees." VQA (Sand dune) — Q: "Where is the man standing?" A: "The man is standing on top of the sand dune."

TABLE V
MILESTONE 4 — GENERATION PARAMETERS BY MODE

Parameter	Single	Paragraph
`max_new_tokens`	50	120
`min_new_tokens`	8	30
`num_beams`	5	5
`no_repeat_ngram_size`	4	4
`length_penalty`	1.0	2.0
`repetition_penalty`	—	2.0

Observations: paragraph mode provides richer context-aware descriptions; single mode is concise and fast; VQA demonstrates strong spatial understanding; slight verbosity observed in paragraph mode, addressed by `repetition_penalty=2.0`. A temperature slider was integrated into the Gradio interface, allowing users to adjust the creativity of generated captions by varying the sampling temperature directly from the UI.

IV. DISCUSSION

LoRA's performance is the most significant finding. With approximately 300K trainable parameters, it outperforms Full Fine-Tuning (124M parameters) on BLEU, METEOR, and ROUGE-L. Full Fine-Tuning's lowest CIDEr score (0.1002) is consistent with overfitting on the small 3,000-sample training set. LoRA's low-rank constraint provides implicit regularization. The result confirms that adapting only the `c_attn` layers with rank-8 matrices is sufficient for this task.

MuseumCaption M3 — Results Summary Table

Model	BLEU	METEOR	CIDEr	ROUGE-L	Trainable Params
Prefix Tuning	0.0450	0.1768	0.1203	0.2601	3.9M
LoRA	0.0537	0.2064	0.1725	0.3028	~300K
Full Fine-tune	0.0474	0.1672	0.1002	0.2598	124M
BLIP	0.0417	0.1937	0.2030	0.3026	247M

Fig. 5. M3 results summary table with exact metric values.

Fig. 6. M3 Test 1 — Soccer scene, all 4 models.

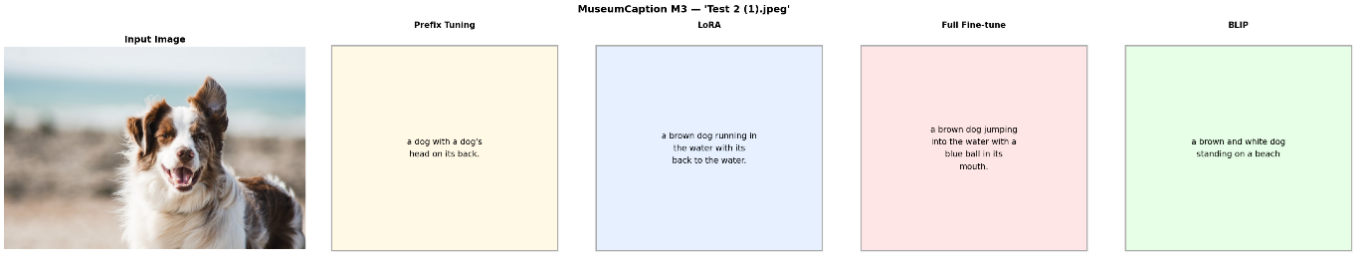

Fig. 7. M3 Test 2 — Dog on beach, all 4 models.

BLIP's CIDEr dominance (0.2030) reflects pretraining data overlap with Flickr30k's distribution. CIDEr weights n-gram frequency against a reference corpus — BLIP's web-scale pretraining provides a natural advantage on this metric. LoRA's superior BLEU, METEOR, and ROUGE-L scores indicate greater lexical diversity. Neither model dominates uniformly, highlighting the importance of multi-metric evaluation.

The beam width and temperature sensitivity experiments reveal that Prefix Tuning degenerates at low beam widths, producing identical outputs at bw=1 and bw=5. Temperature 0.5 collapses to minimal outputs while 1.2 produces semantically incorrect creative outputs. Beam width 5 with nucleus sampling provides the best balance across all GPT-2 variants.

Setting `repetition_penalty=2.0` alongside `length_penalty=2.0` for paragraph mode in Milestone 4 was essential to prevent the BLIP decoder from looping in long-form generation.

Limitations: CIDEr uses simplified TF-IDF cosine similarity rather than original consensus-based CIDEr-D. Evaluation is on 200 samples only. M4 paragraph outputs have no automated metric evaluation.

V. Ethical Reflection

A. *Bias in Training Data*

CLIP and BLIP were pretrained on web-scraped datasets skewing toward Western, English-language content. Flickr30k contains primarily North American and European images. The style prompts designed for museum context risk generating culturally inappropriate descriptions when applied to diverse imagery. Production deployment requires demographic bias auditing.

Fig. 8. M4 Single Caption — Lion. Output: "a lion laying in the grass with a blue sky in the background."

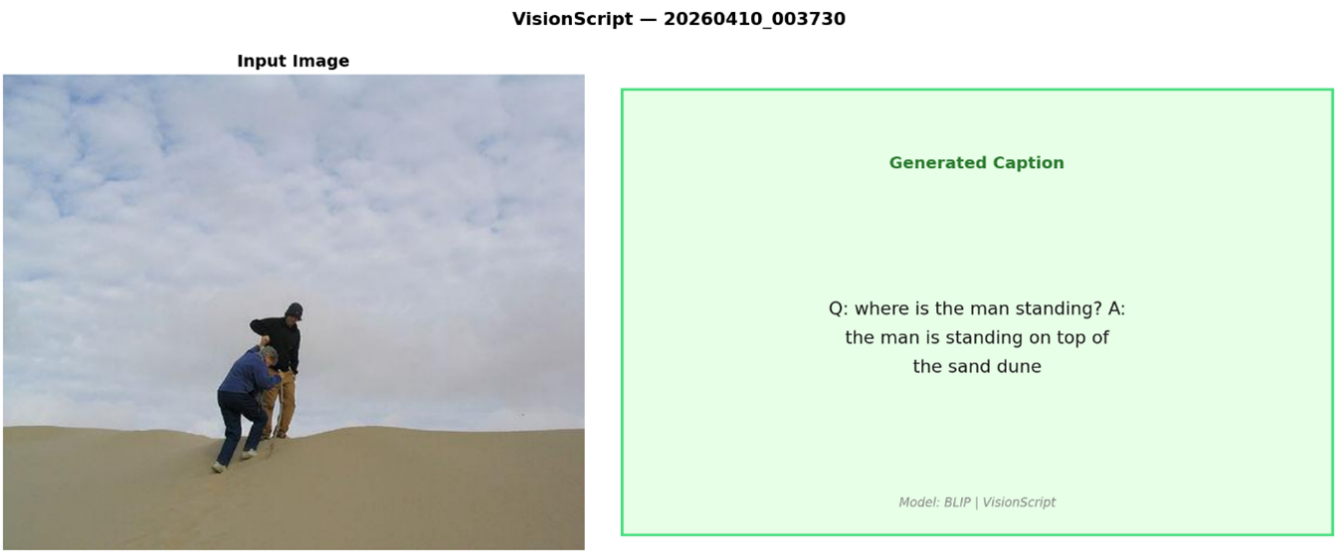

Fig. 9. M4 VQA — Sand dune. Spatial question answered correctly.

B. Privacy and Consent

The VQA module enables targeted queries about individuals' location and appearance. Production deployments must enforce user consent mechanisms and restrict queries about identifiable individuals.

C. Accessibility as a Core Benefit

Paragraph captioning represents genuine value for accessibility. Multi-sentence outputs capturing spatial relationships and scene context substantially exceed standard single-sentence alt text. VQA enables interactive image exploration for visually impaired users.

D. Environmental Cost

LoRA's approximately 400× reduction in gradient computation compared to Full Fine-Tuning makes it the responsible default for academic-scale fine-tuning. Future work should report GPU-hours and estimated CO_2 per experiment.

E. Misuse Potential

A system answering arbitrary questions about image content has dual-use potential for surveillance and unauthorized profiling. Permanent deployment requires rate limiting, content policy enforcement, and terms of service prohibiting surveillance use.

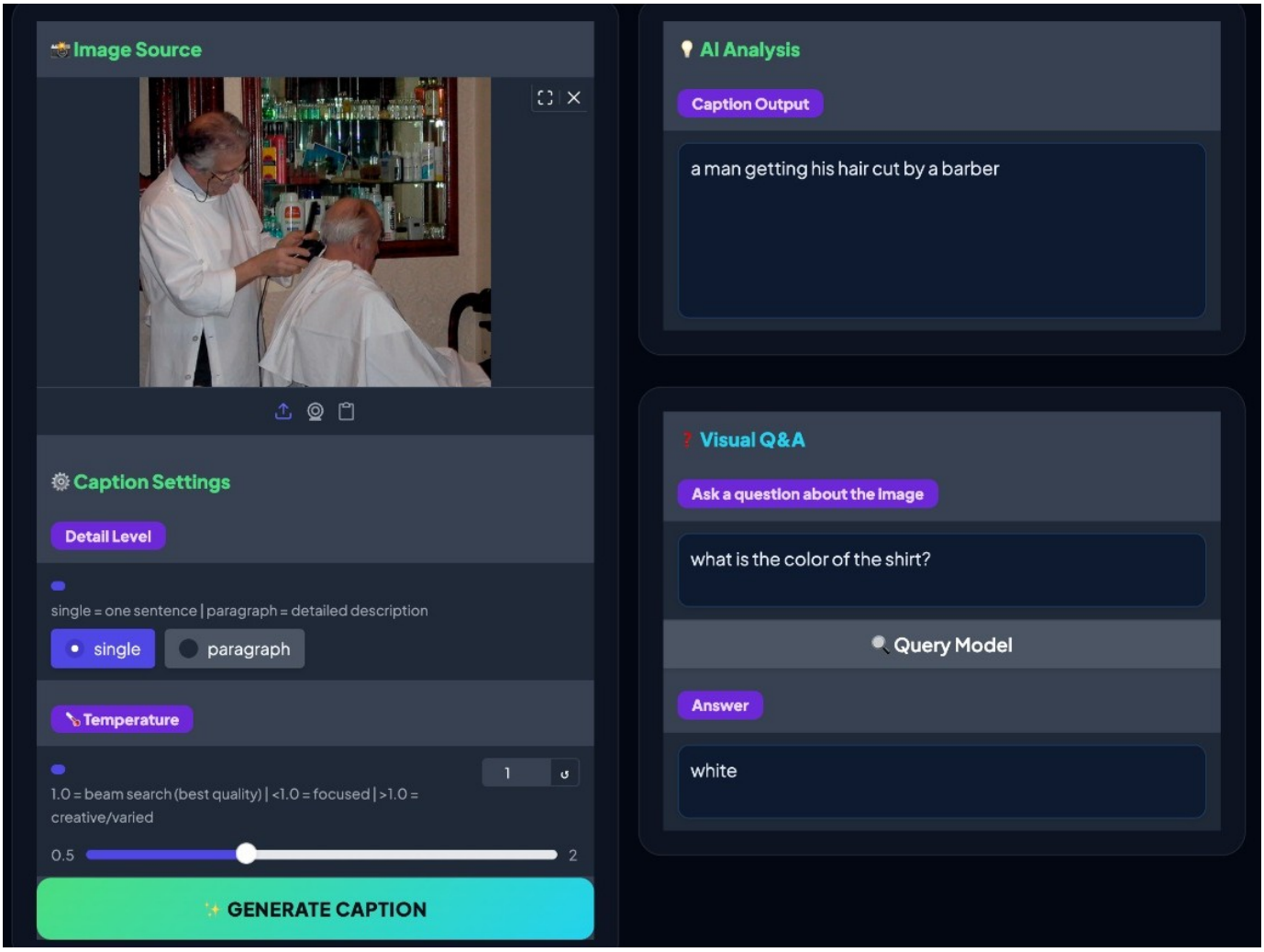

Fig. 10. Milestone 4 — Gradio UI showing single caption mode with temperature slider for creativity control

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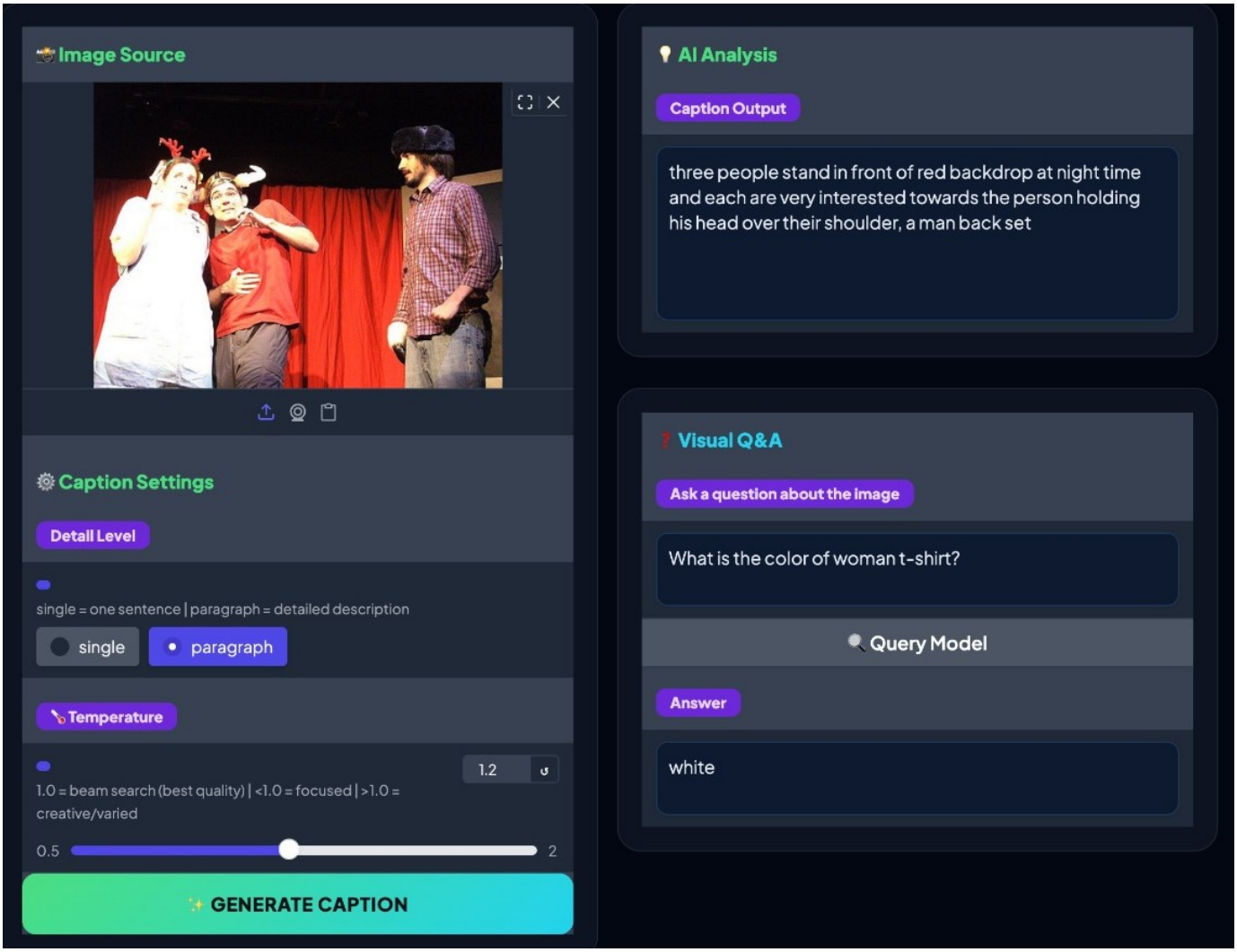

Fig. 11. Milestone 4 — Gradio UI showing paragraph caption mode with temperature slider for creativity control